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Cyber Security Internship Report

Week 03

Module 01 Password Cracking with JTR

Module 02 Password Cracking with JTR using Johnny GUI

Objective:

The objective of these tasks was to understand and practically perform password cracking on a password-protected PDF file using different cybersecurity tools. In Module 01, John the Ripper and Johnny GUI were used to extract and crack the PDF password, while Module 02 used the Networkwalks Hash Calculator and Password Cracker. The activities demonstrated how a password-protected PDF hash can be extracted, analyzed, and used to recover the original password. The tasks also provided practical understanding of weak passwords and the importance of using strong passwords for better security.

Module 1: Password Cracking with JTR

Methodology:

John the Ripper and Johnny GUI were installed and configured on Windows. The PDF hash was extracted and saved in a text file as hash1.txt. The hash was then loaded into Johnny, and a password recovery attack was started using John the Ripper. The password was successfully recovered as password1.

➤ **Step 01: Preparing the Encrypted PDF**

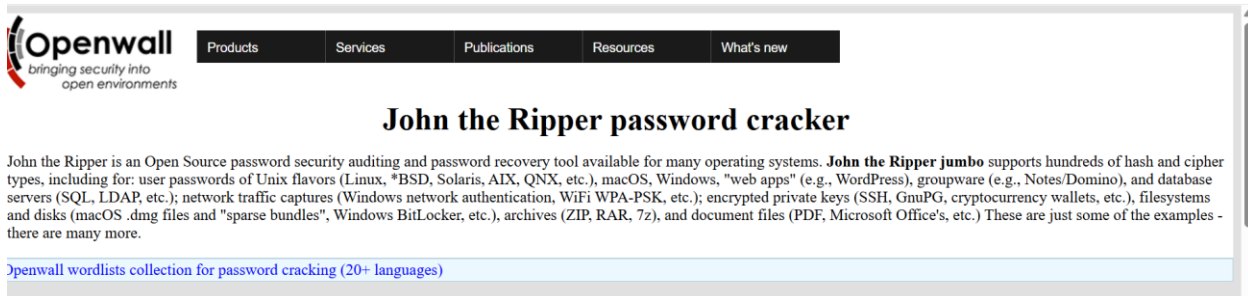
The instructor-provided encrypted PDF file “My Locked PDF1.pdf” was downloaded and placed on the Windows Desktop. This file was used as the authorized lab sample for the password recovery exercise with John the Ripper.



My-Locked-PDF1

➤ Step 02: Downloading John the Ripper

John the Ripper was downloaded from the official Openwall website. The 1.9.0-jumbo-1 64-bit Windows binary ZIP package was selected for the Windows system. The downloaded package contains the John the Ripper executable and required files for password recovery in the authorized lab environment.



- **1.9.0-jumbo-1 64-bit Windows binaries in 7z, 22 MB (signature) or zip, 63 MB (signature)**

```
root@kali: ~
Session Actions Edit View Help
(root@kali)-[~]
# john
Created directory: /root/.john
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100 0
MP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.
```

➤ Step 3: John the Ripper Configuration and Format Verification

```
PS C:\WINDOWS\system32> cd "C:\Users\MT\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run"
>> .\john --help
John the Ripper 1.9.0-jumbo-1 OMP [cygwin 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2019 by Solar Designer and others
Homepage: http://www.openwall.com/john/
```

```
Usage: john [OPTIONS] [PASSWORD-FILES]
--single=SECTION[...] "single crack" mode, using default or named rules
--single=rule[...] same, using "immediate" rule(s)
--wordlist=[FILE] --stdin wordlist mode, read words from FILE or stdin
--pipe like --stdin, but bulk reads, and allows rules
--loopback=[FILE] like --wordlist, but extract words from a .pot file
--dupe-suppression suppress all dupes in wordlist (and force preload)
--prince=[FILE] PRINCE mode, read words from FILE
--encoding=NAME input encoding (eg. UTF-8, ISO-8859-1). See also
doc/ENCODINGS and --list=hidden-options.
--rules=SECTION[...] enable word mangling rules (for wordlist or PRINCE
modes), using default or named rules
--rules=rule[...] same, using "immediate" rule(s)
--rules-stack=SECTION[...] stacked rules, applied after regular rules or to
modes that otherwise don't support rules
--rules-stack=rule[...] same, using "immediate" rule(s)
--incremental=[MODE] "incremental" mode [using section MODE]
--mask=[MASK] mask mode using MASK (or default from john.conf)
--markov=[OPTIONS] "Markov" mode (see doc/MARKOV)
--external=MODE external mode or word filter
--subsets=[CHARSET] "subsets" mode (see doc/SUBSETS)
--stdout=[LENGTH] just output candidate passwords [cut at LENGTH]
--restore=[NAME] restore an interrupted session [called NAME]
--session=NAME give a new session the NAME
--status=[NAME] print status of a session [called NAME]
--make-charset=FILE make a charset file. It will be overwritten
--show=[left] show cracked passwords [if =left, then uncracked]
--test=[TIME] run tests and benchmarks for TIME seconds each
--users=[-]LOGIN[UID[,...]] [do not] load this (these) user(s) only
--groups=[-]GID[,...] load users [not] of this (these) group(s) only
--shells=[-]SHELL[,...] load users with[out] this (these) shell(s) only
```

```
--salts=[-]COUNT[:MAX] load salts with[out] COUNT [to MAX] hashes
--costs=[-]C[:M][,...] load salts with[out] cost value Cn [to Mn]. For
tunable cost parameters, see doc/OPTIONS
--save-memory=LEVEL enable memory saving, at LEVEL 1..3
--node=MIN[-MAX]/TOTAL this node's number range out of TOTAL count
--fork=N fork N processes
--pot=NAME pot file to use
--list=WHAT list capabilities, see --list=help or doc/OPTIONS
--devices=N[,...] set OpenCL device(s) (see --list=opencl-devices)
--format=NAME force hash of type NAME. The supported formats can
be seen with --list=formats and --list=subformats
```

```
PS C:\Users\MT\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> _
```

Successful execution of John the Ripper help command

```
PS C:\Users\MT\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john --list=formats
bcrypt, bsdcrypt, md5crypt, md5crypt-long, bcrypt, scrypt, LM, AFS,
tripcode, AndroidBackup, adxcrypt, agilekeychain, aix-ssh1, aix-ssh256,
aix-ssh512, andOTP, ansible, argon2, as400-des, as400-ssh1, asa-md5,
AxCrypt, AzureAD, BestCrypt, bfeegg, Bitcoin, BitLocker, bitshares, Bitwarden,
BKS, BlackBerry-ES10, WoWSRP, Blockchain, chap, Clipperz, cloudkeychain,
dynamic_n, cq, CRC32, sha1crypt, sha256crypt, sha512crypt, Citrix_NS10,
dahua, dashlane, diskcryptor, Django, django-scrypt, dmd5, dmg, dominosec,
dominosec8, DPAPImk, dragonfly3-32, dragonfly3-64, dragonfly4-32,
dragonfly4-64, Drupal7, eCryptfs, eigrp, electrum, EncFS, enpass, EPI,
EPiServer, ethereum, fde, Fortigate256, Fortigate, FormSpring, FVDE, geli,
gost, gpg, HAVAL-128-4, HAVAL-256-3, hdaa, hMailServer, hsrp, IKE, ipb2,
itunes-backup, iwork, KeePass, keychain, keyring, keystore, known_hosts,
krb4, krb5, krb5asrep, krb5pa-sha1, krb5tgs, krb5-17, krb5-18, krb5-3,
kwallet, lp, lpcii, leet, lotus5, lotus85, LUKS, MD2, mdc2, MediaWiki,
monero, money, MongoDB, scram, Mozilla, mscash, mscash2, MSCHAPv2,
mschapv2-naive, krb5pa-md5, mssql, mssql05, mssql12, multibit, mysqlna,
mysql-sha1, mysql, net-ah, nethalflm, netlm, netlmv2, net-md5, netntlmv2,
netntlm, netntlm-naive, net-sha1, nk, notes, md5ns, nsec3, NT, o10glogon,
o3logon, o5logon, ODF, Office, oldoffice, OpenBSD-SoftRAID, openssl-enc,
oracle, oracle11, Oracle12C, osc, ospf, Padlock, Palshop, Panama,
PBKDF2-HMAC-MD4, PBKDF2-HMAC-MD5, PBKDF2-HMAC-SHA1, PBKDF2-HMAC-SHA256,
PBKDF2-HMAC-SHA512, PDF, PEM, pfx, pgpdisk, pgpsda, pgpwde, phpass, PHPS,
PHPS2, pix-md5, PKZIP, po, postgres, PST, PuTTY, pwsafe, qnx, RACF,
RACF-KDFAES, radius, RAdmin, RAKP, rar, RAR5, Raw-SHA512, Raw-Blake2,
Raw-Keccak, Raw-Keccak-256, Raw-MD4, Raw-MD5, Raw-MD5u, Raw-SHA1,
Raw-SHA1-AxCrypt, Raw-SHA1-LinkedIn, Raw-SHA224, Raw-SHA256, Raw-SHA3,
Raw-SHA384, ripemd-128, ripemd-160, rsvp, Siemens-S7, Salted-SHA1, SSHA512,
sappb, sappg, saph, sappse, securezip, 7z, Signal, SIP, skein-256, skein-512,
skey, SL3, Snefru-128, Snefru-256, LastPass, SNMP, solarwinds, SSH, ssrp,
Stribog-256, Stribog-512, STRIP, SunMD5, SybaseASE, Sybase-PROP, tacacs-plus,
```

```
tc_sha512, tc_whirlpool, vdi, OpenVMS, vmx, VNC, vtp, wbb3, whirlpool,
whirlpool0, whirlpool1, wpapsk, wpapsk-pmk, xmpp-scram, xsha, xsha512, ZIP,
ZipMonster, plaintext, has-160, HMAC-MD5, HMAC-SHA1, HMAC-SHA224,
HMAC-SHA256, HMAC-SHA384, HMAC-SHA512, sha1crypt-openc1, KeePass-openc1,
oldoffice-openc1, PBKDF2-HMAC-MD4-openc1, PBKDF2-HMAC-MD5-openc1,
PBKDF2-HMAC-SHA1-openc1, rar-openc1, RAR5-openc1, TrueCrypt-openc1,
lotus5-openc1, AndroidBackup-openc1, agilekeychain-openc1, ansible-openc1,
axcrypt-openc1, axcrypt2-openc1, bcrypt-openc1, BitLocker-openc1,
bitwarden-openc1, blockchain-openc1, cloudkeychain-openc1, md5crypt-openc1,
sha256crypt-openc1, sha512crypt-openc1, dashlane-openc1, decrypt-openc1,
diskcryptor-openc1, diskcryptor-aes-openc1, dmg-openc1,
electrum-modern-openc1, EncFS-openc1, enpass-openc1, ethereum-openc1,
ethereum-presale-openc1, FVDE-openc1, geli-openc1, gpg-openc1, iwork-openc1,
keychain-openc1, keyring-openc1, keystore-openc1, krb5pa-md5-openc1,
krb5pa-sha1-openc1, krb5asrep-aes-openc1, lp-openc1, lpc1i-openc1, LM-openc1,
mscash-openc1, mscash2-openc1, mysql-sha1-openc1, notes-openc1, NT-openc1,
ntlmv2-openc1, o5logon-openc1, ODF-openc1, office-openc1,
OpenBSD-SoftRAID-openc1, PBKDF2-HMAC-SHA256-openc1,
PBKDF2-HMAC-SHA512-openc1, pem-openc1, pfx-openc1, pgpdisk-openc1,
pgpsda-openc1, pgpwde-openc1, PHPass-openc1, pwsafe-openc1, RAKP-openc1,
raw-MD4-openc1, raw-MD5-openc1, raw-SHA1-openc1, raw-SHA256-openc1,
raw-SHA512-free-openc1, raw-SHA512-openc1, salted-SHA1-openc1, sappse-openc1,
7z-openc1, SL3-openc1, solarwinds-openc1, ssh-openc1, sspr-openc1,
strip-openc1, telegram-openc1, tezos-openc1, vmx-openc1, wpapsk-openc1,
wpapsk-pmk-openc1, XSHA512-free-openc1, XSHA512-openc1, ZIP-openc1, dummy,
crypt
PS C:\Users\MT\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run>
```

Select and Upload your PDF:

[Remove](#) [Upload](#) [Browse ...](#)

Remove

Upload

[Browse ...](#)

[illegible][illegible]

Extracted PDF Hash Value



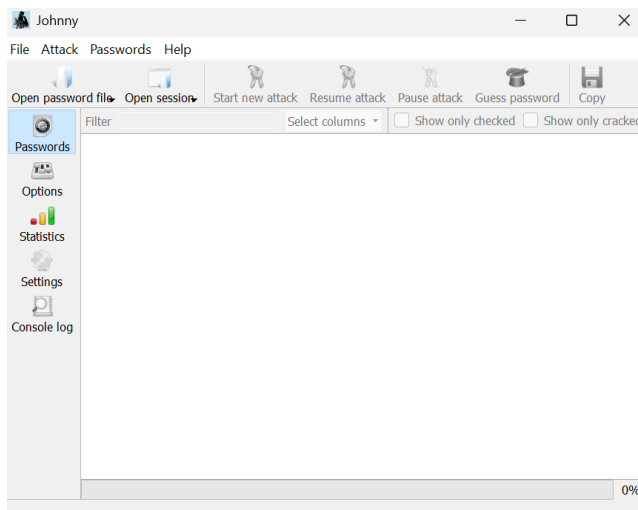
Johnny Setup

Completing the Johnny Wizard

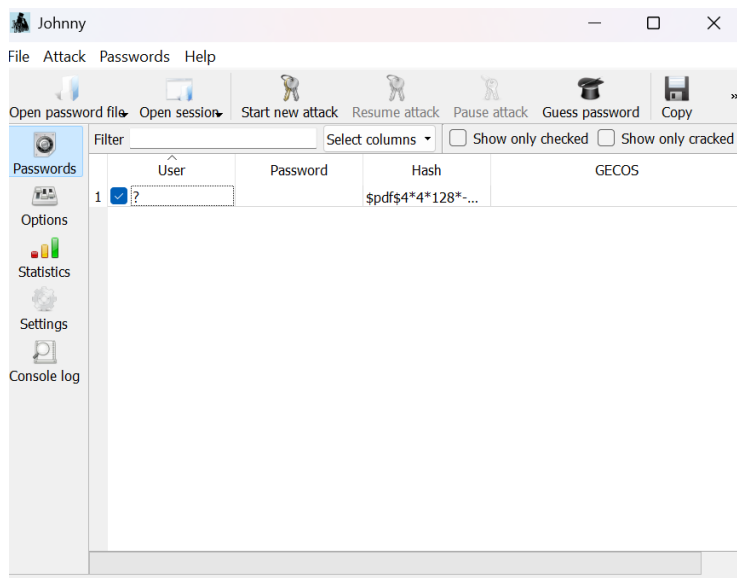
Click Finish to exit the Johnny Wizard.

☒ Run Johnny now.

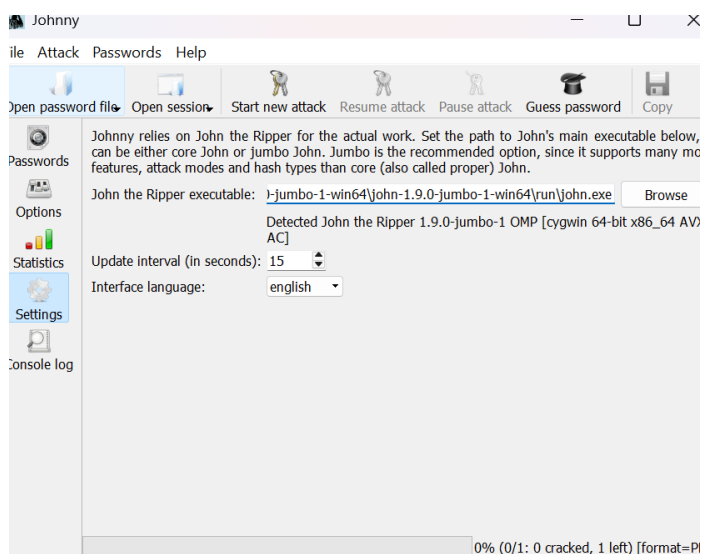
Finish



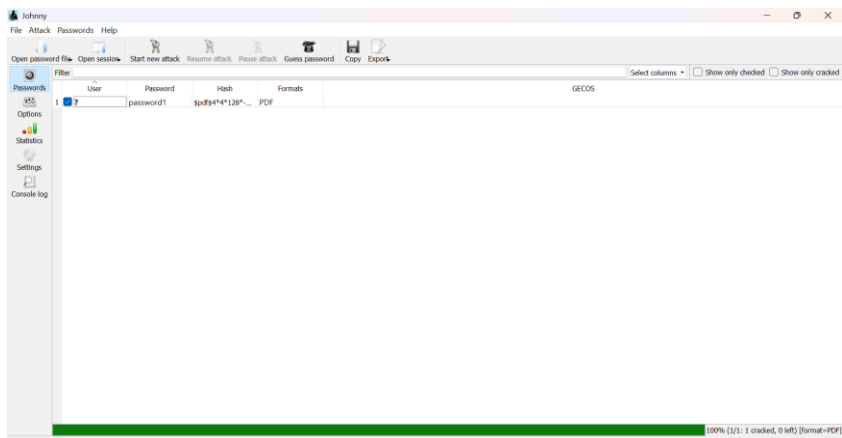
Johnny GUI Installation Setup



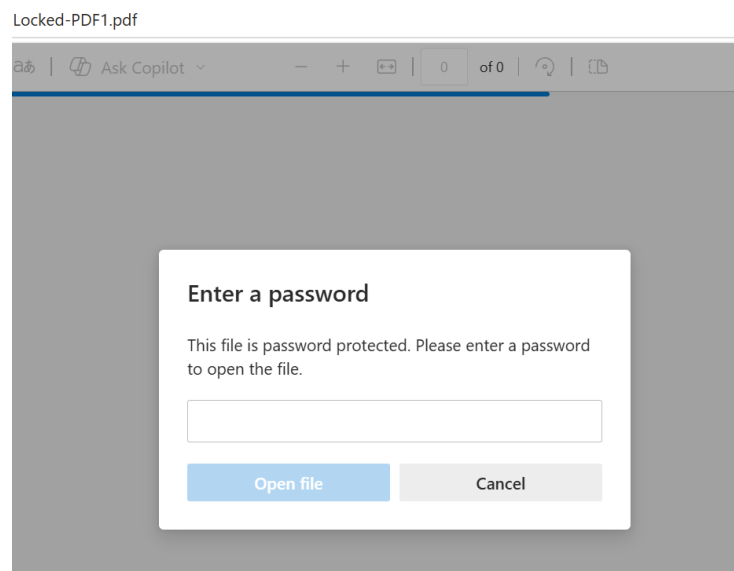
Loading the PDF Hash File in Johnny



Johnny Settings with John the Ripper Executable Path

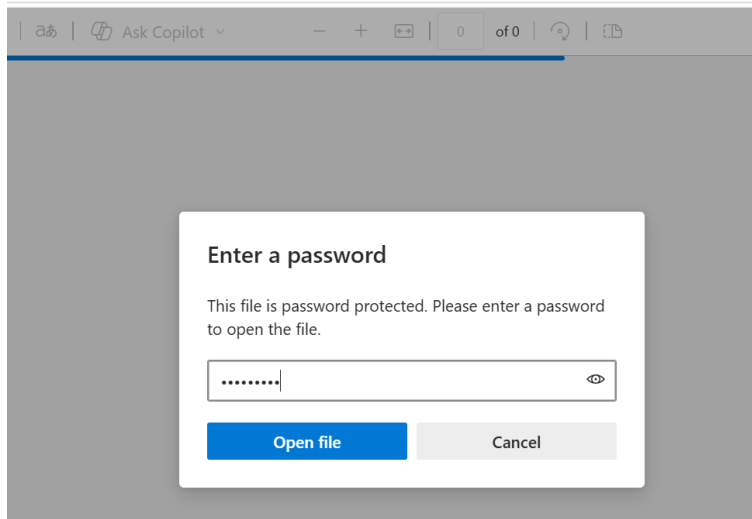


PDF Hash Loaded in Johnny Before Password Recovery Attack

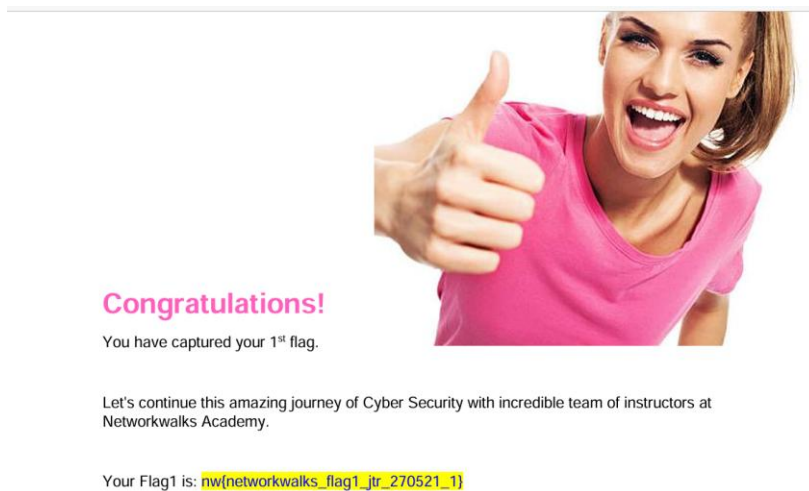


PDF Password Prompt

My-Locked-PDF1.pdf



Locked PDF File and Hash File



Successfully Unlocked PDF and Captured Flag

Module 02: Password Cracking With Networking tools

Methodology:

The same encrypted PDF was processed using the Networkwalks Hash Calculator. The PDF hash was extracted and copied into the Networkwalks Password Cracker. The password-cracking process was started, and the password password1 was

successfully recovered. Finally, the password was entered into the protected PDF, which opened successfully and displayed the first flag.

➤ **STEP 1: Download the Encrypted PDF File**



My-Locked-PDF1

➤ **STEP 2: Open the Network walks Hash Calculator**



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



Text

File

PDF

Enter text to hash

Type or paste text. Hashes update as you type.

☒ **Uppercase output**

Every hash is computed locally in your browser. SHA hashes use the built in Web Crypto API and MD5 uses an in page implementation. No text or file is ever uploaded. Note that MD5 and SHA-1 are broken for security use and are fine only for checksums and integrity checks.

➤ STEP 3: Upload the Locked PDF File

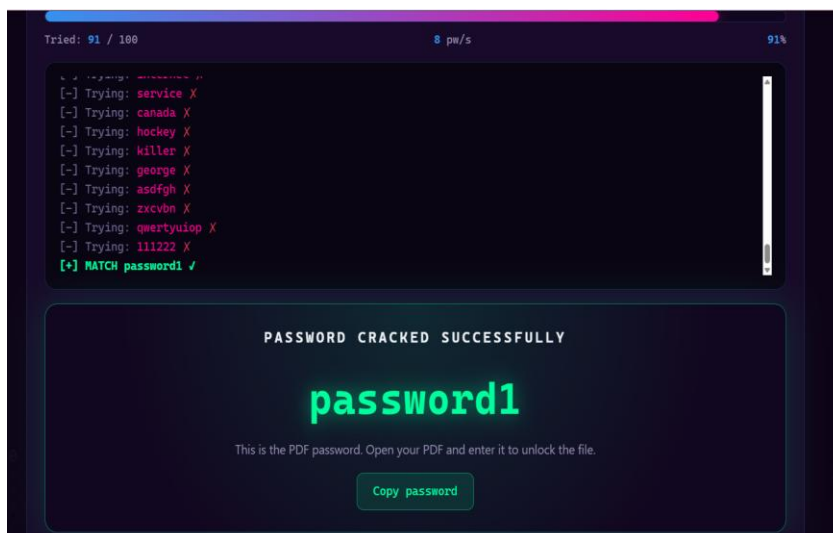
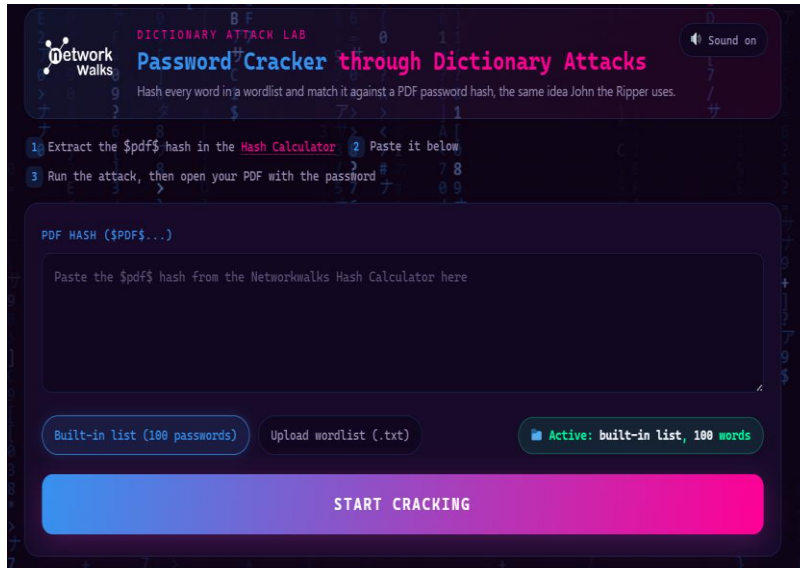
```
My-Locked-PDF1.pdf | 65.2 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash
$pdf$4*4*128*-1060*1*16*55d1a5c14175da449753199e44971d32*32*777fd021a7f3c5ae59
8c0c6495c7f76e0000000000000000000000000000000000000000000000000000*32*ceecdac74b19b5a62688d3b3524e
1374c955cbb9cc3c45316494d9446ef81af1

Revision: R4 Version: V4 Key length: 28 bit
```

➤ STEP 4: Crack the PDF Password Using Network walks Password Crack



➤ STEP 5: Enter the Cracked Password in the PDF



Congratulations!

You have captured your 1st flag.

Let's continue this amazing journey of Cyber Security with incredible team of instructors at Networkwalks Academy.

Your Flag1 is: `nw(networkwalks_flag1_jtr_270521_1)`

Conclusion:

Both modules were successfully completed. The password-protected PDF was processed using John the Ripper and Johnny GUI in Module 01 and the Networkwalks Hash Calculator and Password Cracker in Module 02. The password was successfully recovered as password1, and the protected PDF was opened successfully to capture the first flag. These practical tasks improved understanding of password cracking, PDF hash extraction, and password security. The activities also demonstrated why strong and unique passwords are important for protecting files and sensitive information.